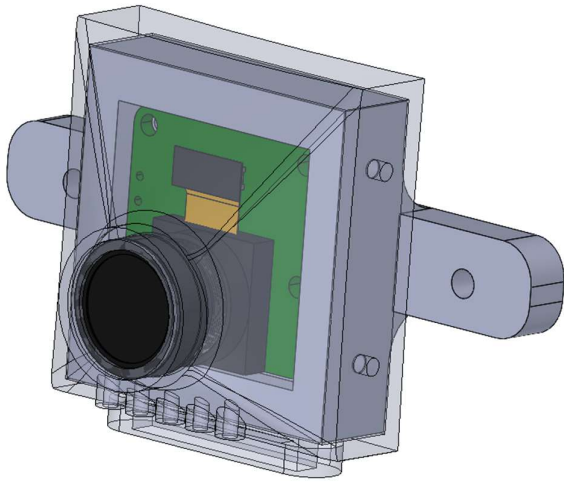


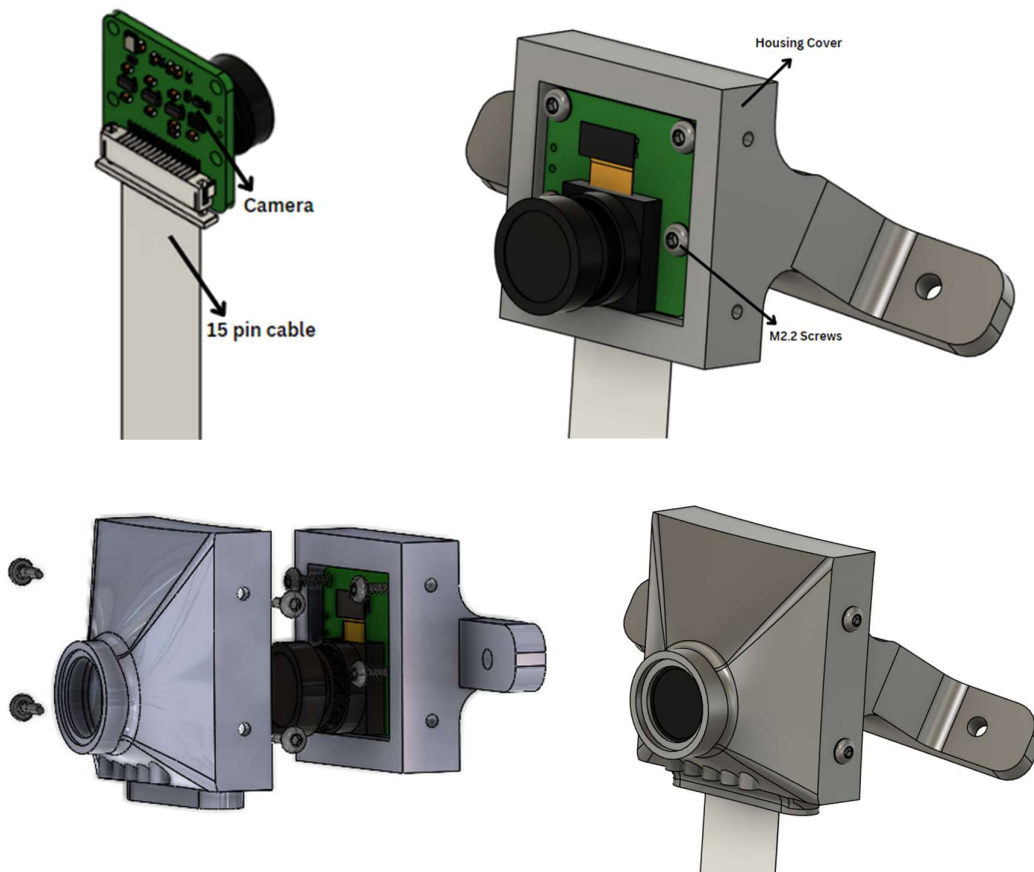


UPWORK YUSUF PROJECT

Cameras

There are two parts to the housing for Camera housing. One part attaches to the pole and has screws for mounting the camera. The other part is the front cover, which protects the camera module. This cover includes air vents at the bottom and can be attached and detached using screws. The housing is designed to withstand rainwater.

The first step is to connect the 15-pin wire to the camera, then pass the wire through the slot and secure the camera with screws. After that, attach the top cover and fix it in place with screws. This process is the same for all cameras. Finally, the camera is mounted to the pole using nuts and bolts.



M2.2 Screws: <https://www.amazon.com/uxcell-M2-2x6-5mm-Phillips-Tapping-Screws/dp/B01EHP3HLC>

CSI 15-15 pin long wires: <https://www.amazon.com/Aokin-Raspberry-Camera-Flexible-Ribbon/dp/B07WF91QML?th=1>

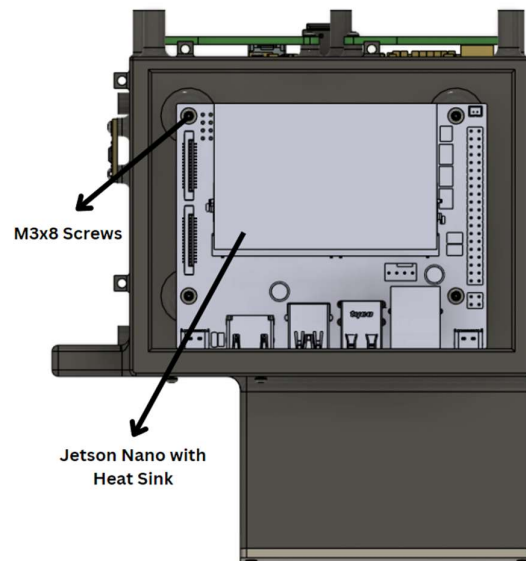
Electronics Components Housing

The electronics housing consists of two parts: an inner/main housing that holds the electronic components, and an outer cover that protects everything from rain.

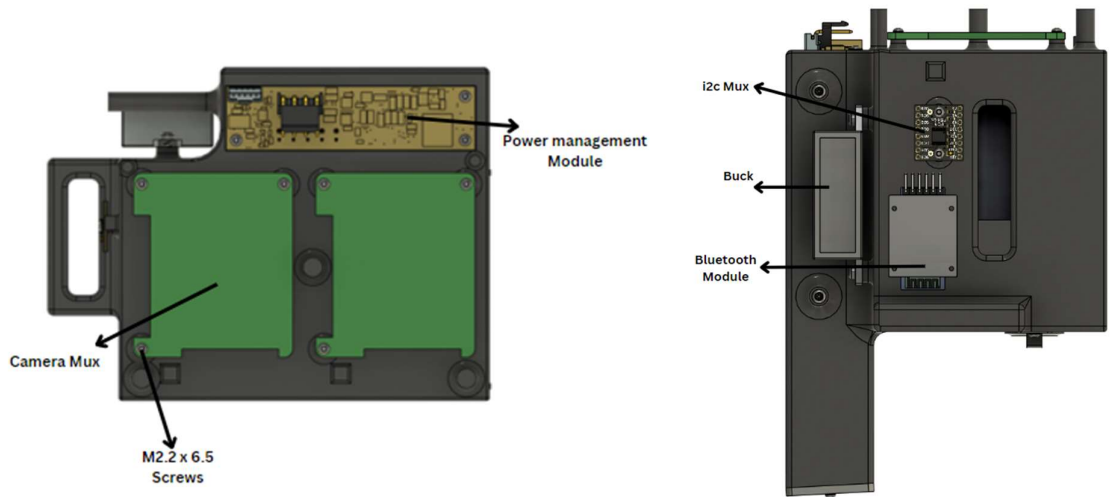


Inner Housing

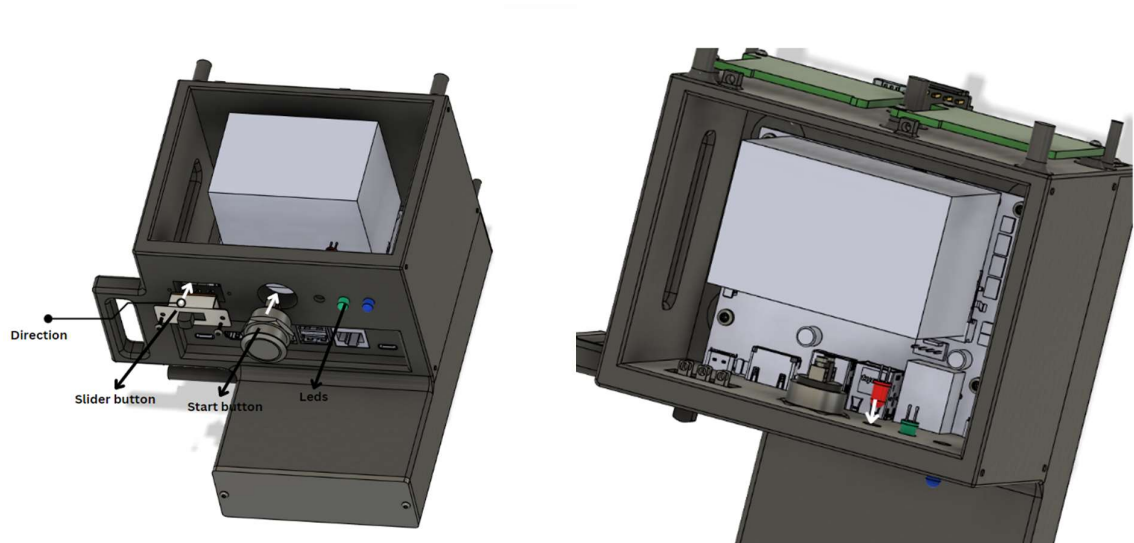
The Jetson Nano will be secured first with M3x8 screws. On top of it, the camera multiplexer and Power Management Module (RRC-PMM20) will be mounted, with an I2C multiplexer fixed on the side using M2.2x6.5 screws (the same used in the camera housings), as illustrated in the picture.



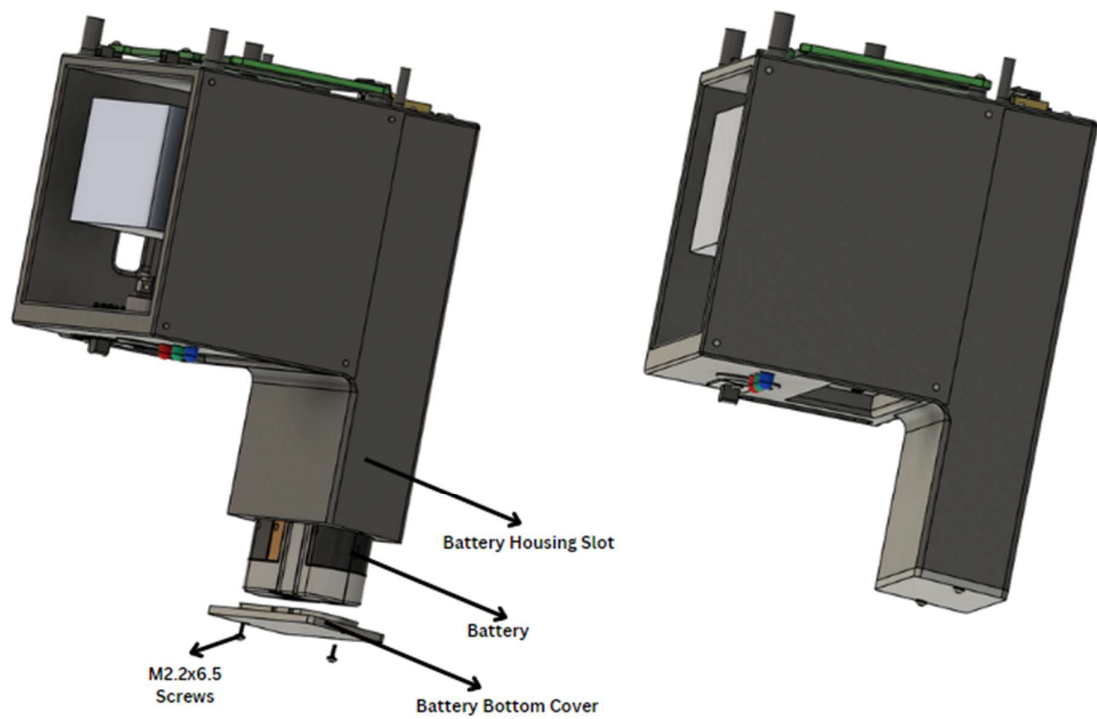
The housing design allows the battery to be easily removed without needing to completely dismantle the housing. A bottom cover, attached with M2.2x6 screws, will hold the battery in place.



The LEDs will be installed from the inner side of the housing, while the start button and slider will be mounted externally, with the wiring pre-soldered before installation.

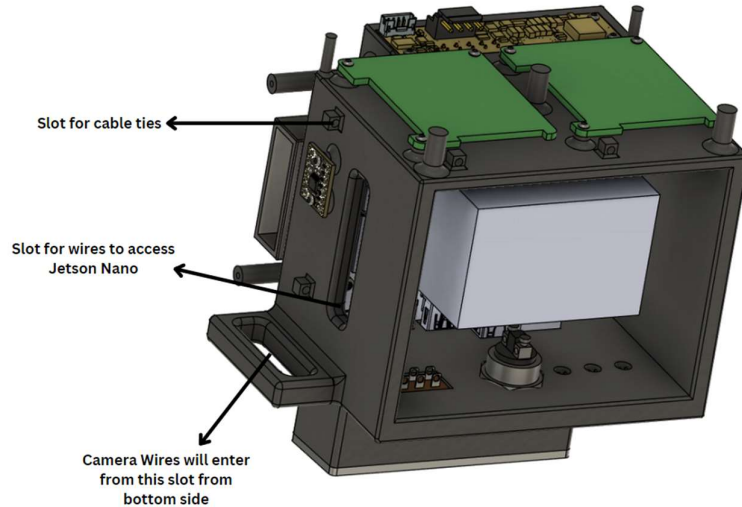


The battery slides in from the bottom and locks into place with the power management module connector. It is secured with a bottom cover using screws, allowing for easy removal.

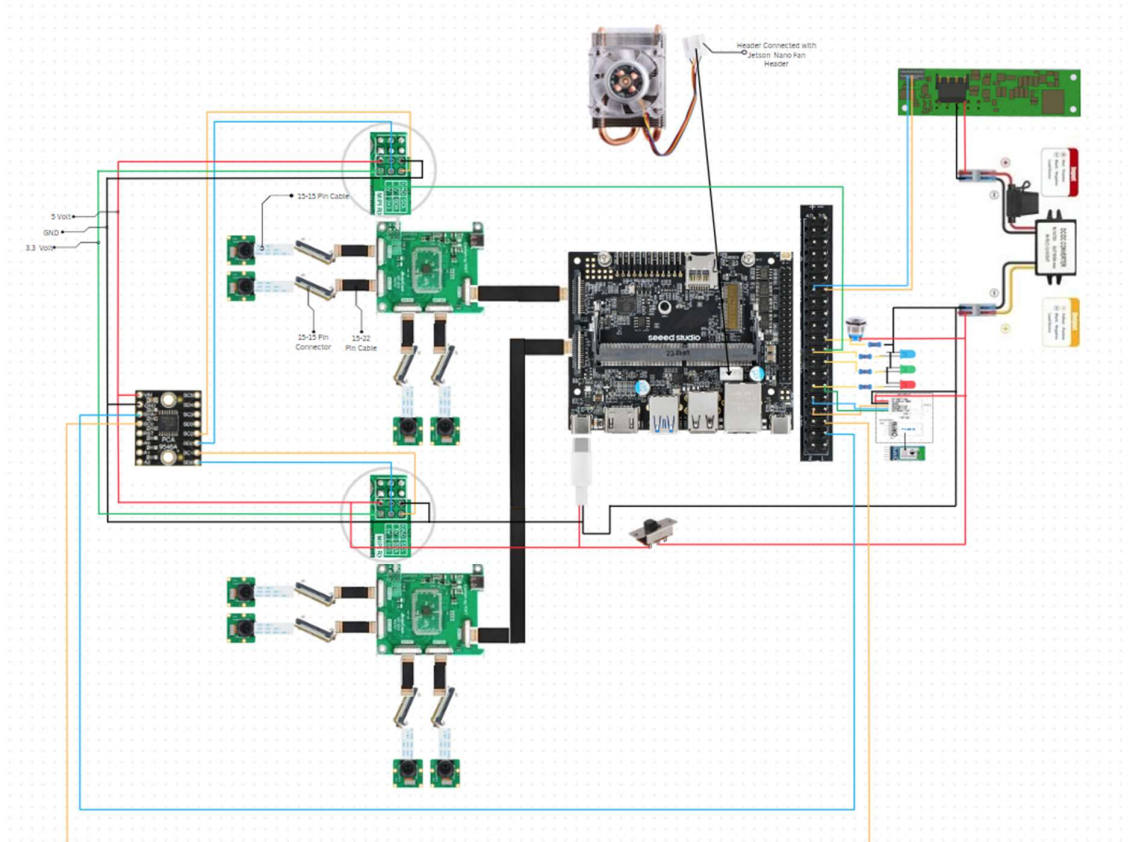


Wiring:

The camera wires (15-pin, long) will enter through the bottom slot (11x45 mm), providing ample space for them. Additionally, a slot (12x55 mm) is provided for access to the Raspberry Pi. Small hooks are included for cable management, allowing the use of cable ties to keep everything organized.



This is the connection diagram. For high resolution, please open the “circuit.pdf” file. The connections follow the pinout provided by the client, except for the camera multiplexers. Both camera multiplexers are connected to an I2C multiplexer. The SDA wire from the I2C multiplexer is connected to Pin 3, and the SCL wire is connected to Pin 5.



PMM20 Power Module:

The PMM20 module pinout is provided below. For more detailed information, please refer to the PMM20.pdf document.

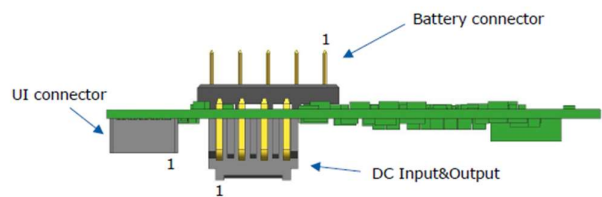
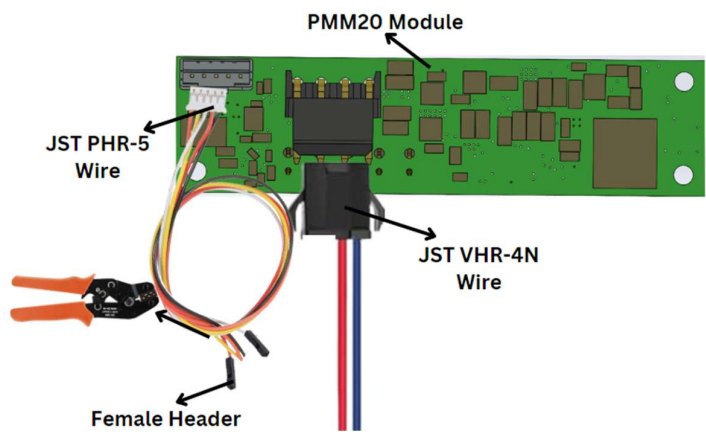


Figure 1 – Connectors sequence of PMM20. Pin 1 of each connector is designated by a small “1”.

Connector	Description
DC Input & Output	This connector functions as both the input and output power connection.
	Pin 1: V_{out} (connects to the application)
	Pin 2: GND
	Pin 3: GND
	Pin 4: V_{in} (connects to an external power supply)
User Interface (UI)	Pin 1: SMBus Data (SDA) Data signal for SMBus.
	Pin 2: SMBus Clock (SCL) Clock signal for SMBus.
	Pin 3: NRST Pull to GND to reset the PMM20.
	Pin 4: LED Indicator Connect an LED for outputting status information.
	Pin 5: GND Ground pin.

We will use the recommended connector for DC input and output, utilizing only two wires since we only need to take output from the battery, and the 22 AWG wires can easily handle the current. The unnecessary wires will be cut. The UI connector will be used for communication, but since it doesn’t come with direct female headers to connect with the Jetson Nano, we’ll crimp female headers using a crimping tool only on the wires that will connect to the Jetson Nano.



DC Input and Output: <https://www.amazon.de/-/en/Lyeteung-Pairs-Connectors-Micro-Female-4-pole/dp/B0CBWY3G45?th=1>

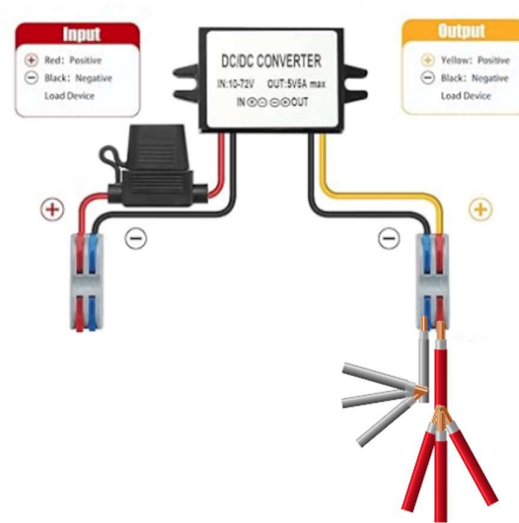
UI Connector: <https://www.amazon.de/-/en/SGerste-Mini-Micro-Plug-Cables/dp/B078TZ5PNY>

Crimping tool: <https://a.co/d/2dkuDHu>

Female Headers: <https://a.co/d/hAJn5DL>

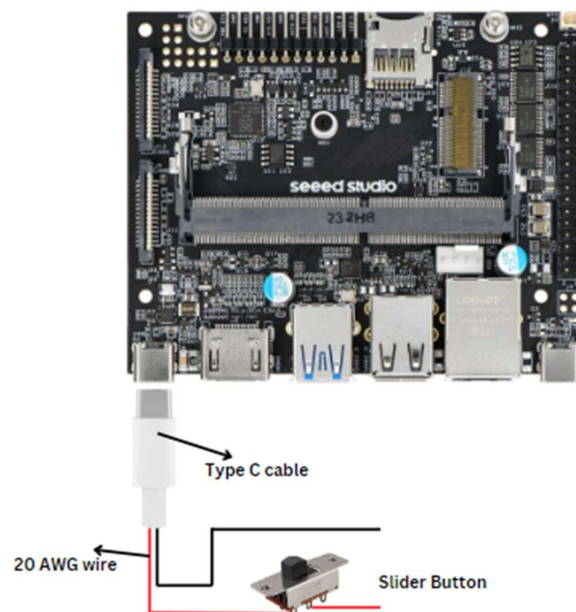
BUCK:

The buck already has wires and the connectors attached to it. The 2 wires from the battery will attach to the input side and on the output side all the 5 volt and ground wires will come and connected as show below.



Jetson Nano Power:

The Jetson Nano is powered by a USB cable. The red wire from the USB cable is soldered to the slider button and then connected to the buck converter's output. The ground wire is also soldered and connected to the buck converter's output ground.



The 20 AWG wire is more than sufficient for the current required and can easily handle the load.

Quick Calculator

Wire Length (feet)

0.5

Wire Gauge (AWG)

20

CALCULATE

Wire Resistance

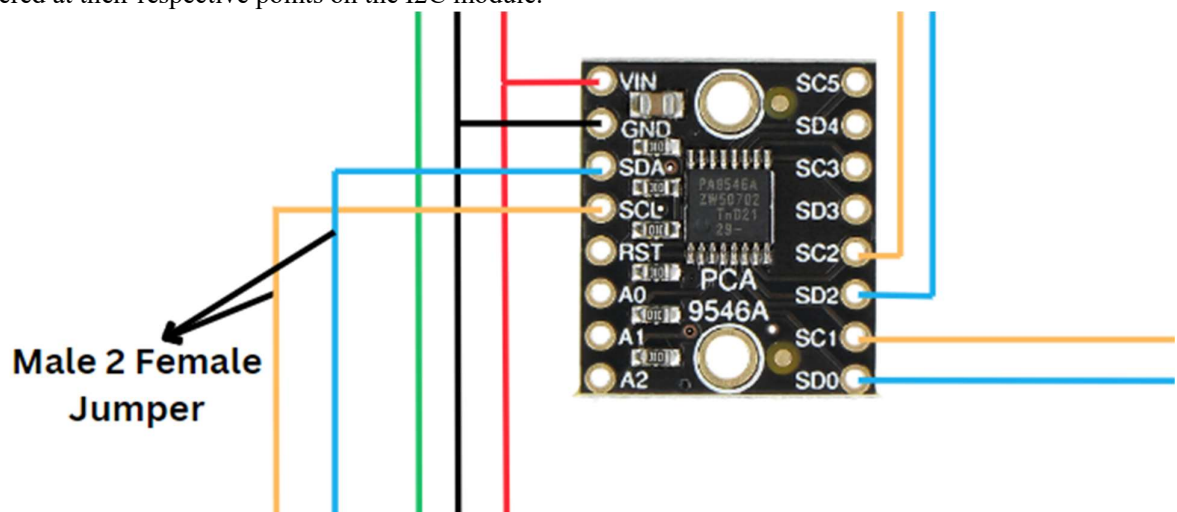
0.005 ohms (Results are rounded to the nearest milliohm.)

Type C Cable: <https://a.co/d/6Rj53DL>

20 AWG wires: <https://a.co/d/5hOcLr8>

I2C Module:

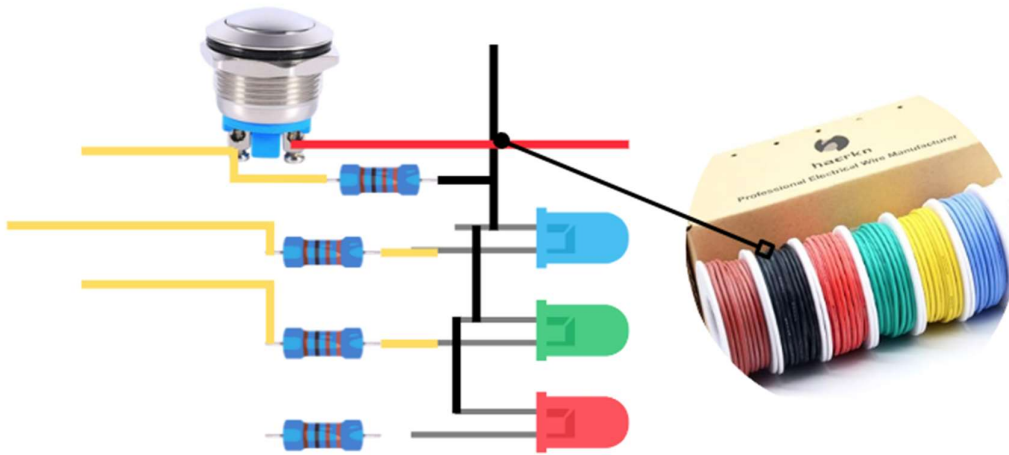
The I2C module is connected using 22 AWG jumper wires. The Jetson Nano wires are connected with male-to-female jumpers, while all the other connections use male-to-male jumpers. The male-to-male jumpers are soldered at their respective points on the I2C module.



Jumper: <https://a.co/d/atIbyK4>

LEDS and Buttons:

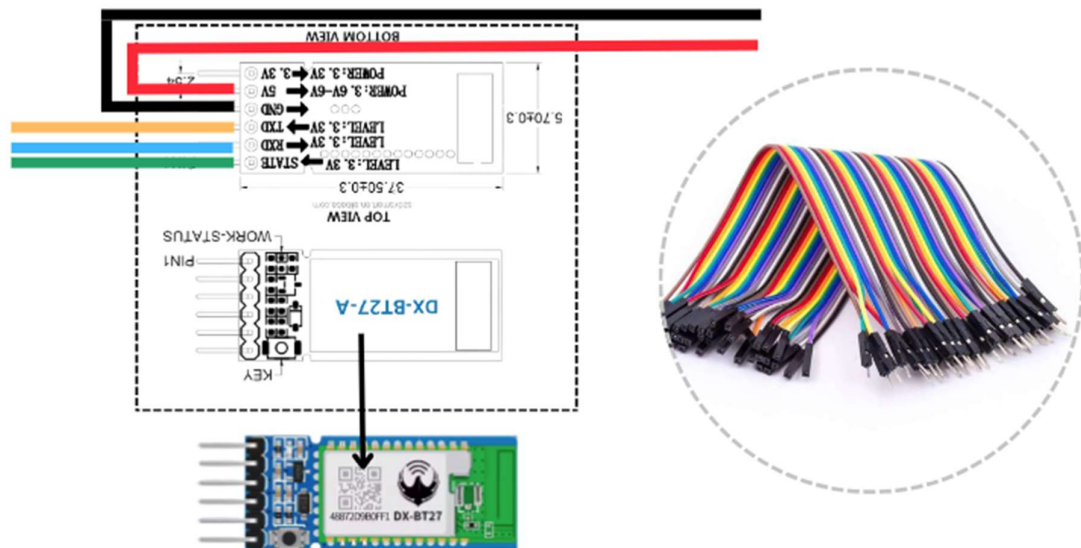
The LEDs and buttons are soldered using the same 20 AWG wire, ensuring a reliable connection that can easily handle the required current.



20 AWG wires: <https://a.co/d/5hOcLr8>

Bluetooth Module:

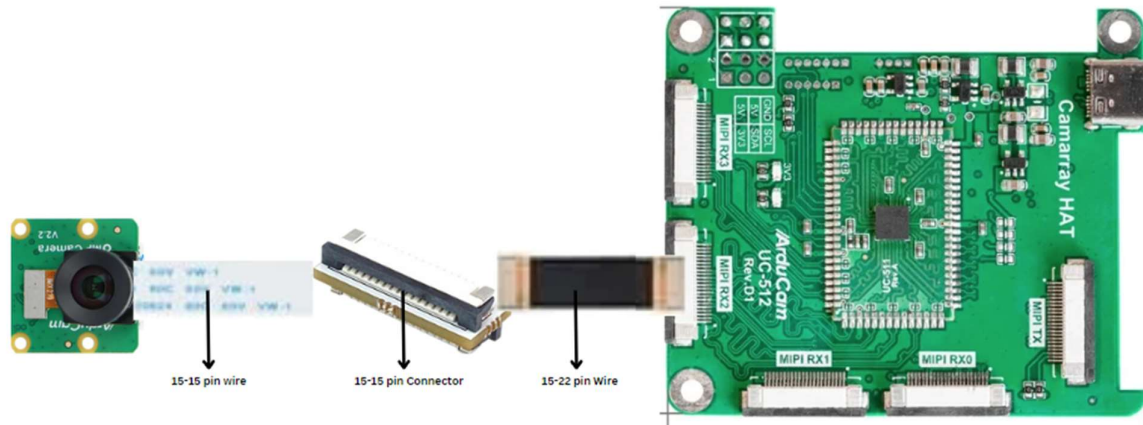
The Bluetooth module is connected using the female to female jumpers.



Jumpers: <https://a.co/d/atIbyK4>

Cameras:

First, a 15-pin to 15-pin wire is connected to the camera. These wires are chosen because they are available in longer lengths, whereas 15-pin to 22-pin wires are not commonly available in long lengths. The 15-pin to 15-pin wire connects to a 15-pin to 15-pin connector, and then a 15-pin to 22-pin wire is used to connect to the mux, as the mux has 22-pin connectors.



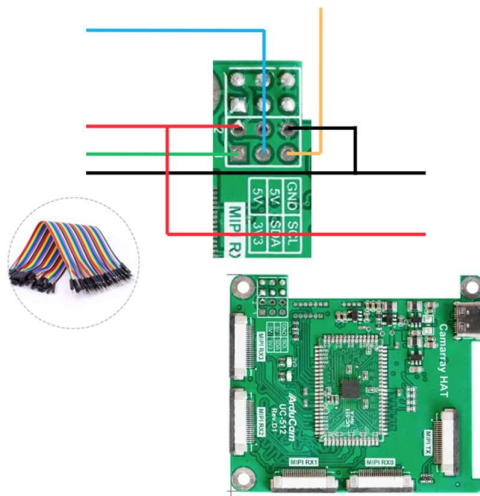
15-15 pin wire: <https://www.amazon.com/Aokin-Raspberry-Camera-Flexible-Ribbon/dp/B07WF91QML?th=1>

15-15 pin connector: <https://www.reichelt.com/de/en/raspberry-picamera-cable-connector-15-pin-to-15-pinrpi-cam-15-15-p319044.html>

15-22 pin wire: <https://www.amazon.com/Flexible-Raspberry-Suitable-Modules-XYGStudy/dp/B0CNP5RJMT?th=1>

Camera Mux:

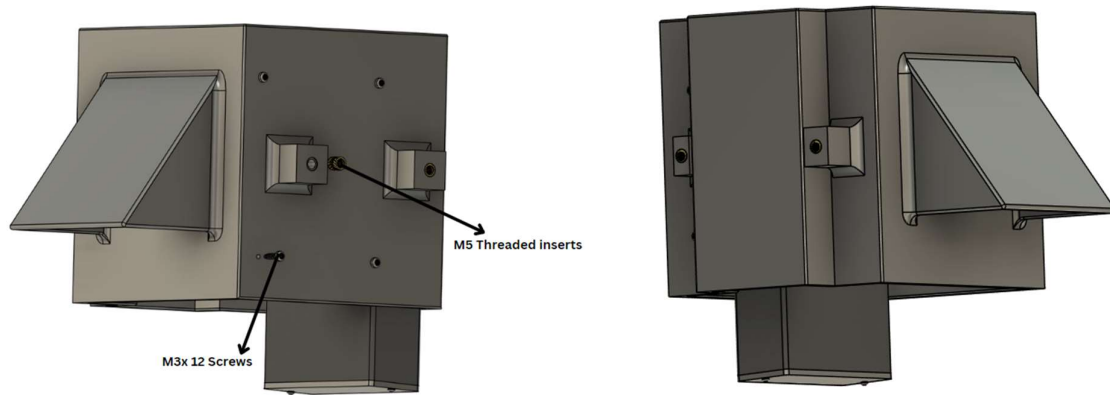
The camera mux uses male-to-female jumpers, with the male jumpers soldered to the mux and the female jumpers connected to the Jetson Nano.



Jumpers: <https://a.co/d/atIbyK4>

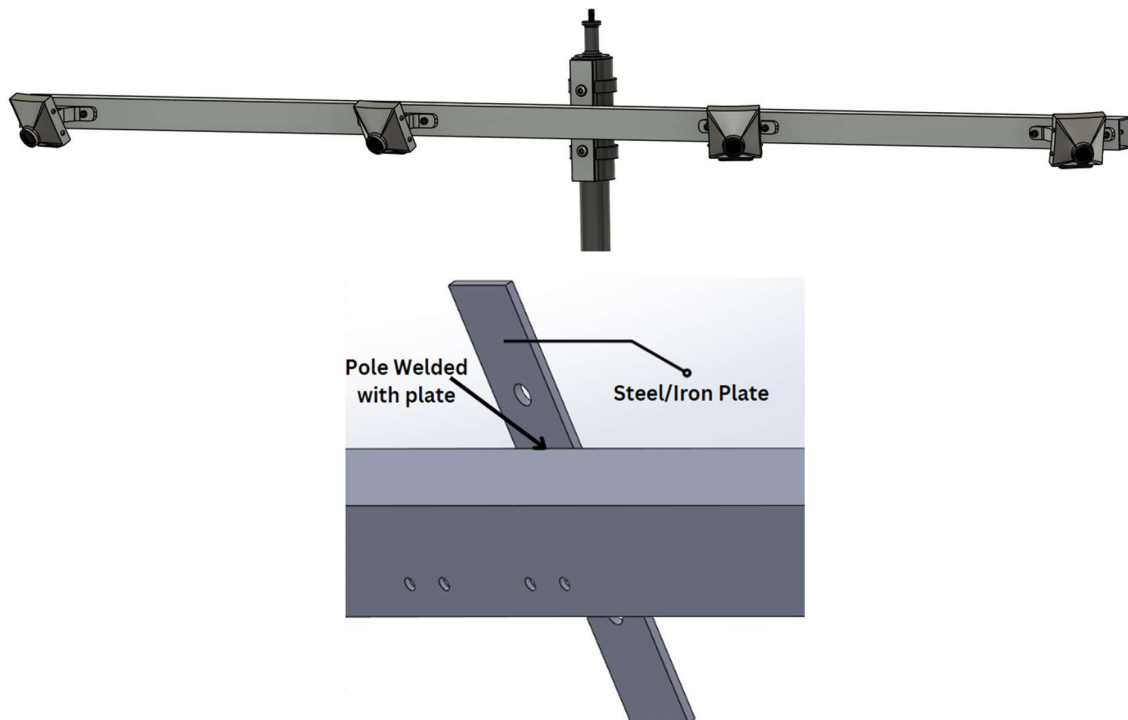
Outer Housing,

After securing and connecting all the components, the housing cover will be placed on top to enclose everything. The design ensures that rainwater will not affect the internal components.

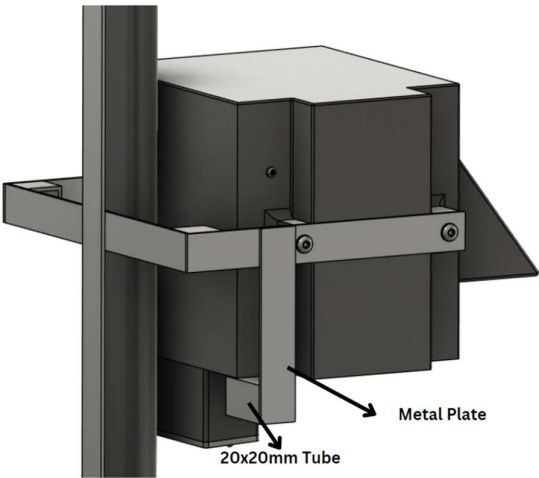


Top Pole 1 meter

The poles are made up metal which are welded together and have holes to secure components to it.

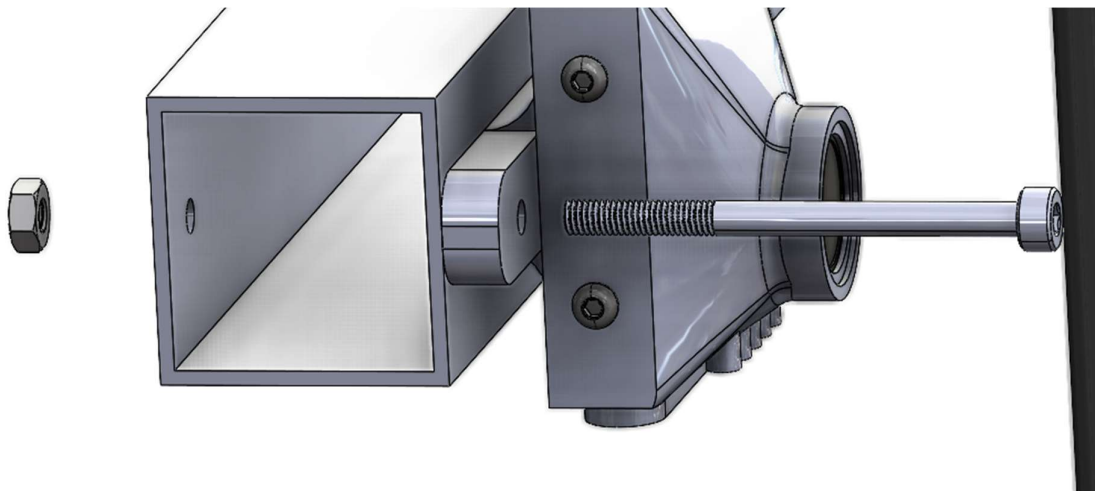


1.5-meter Pole

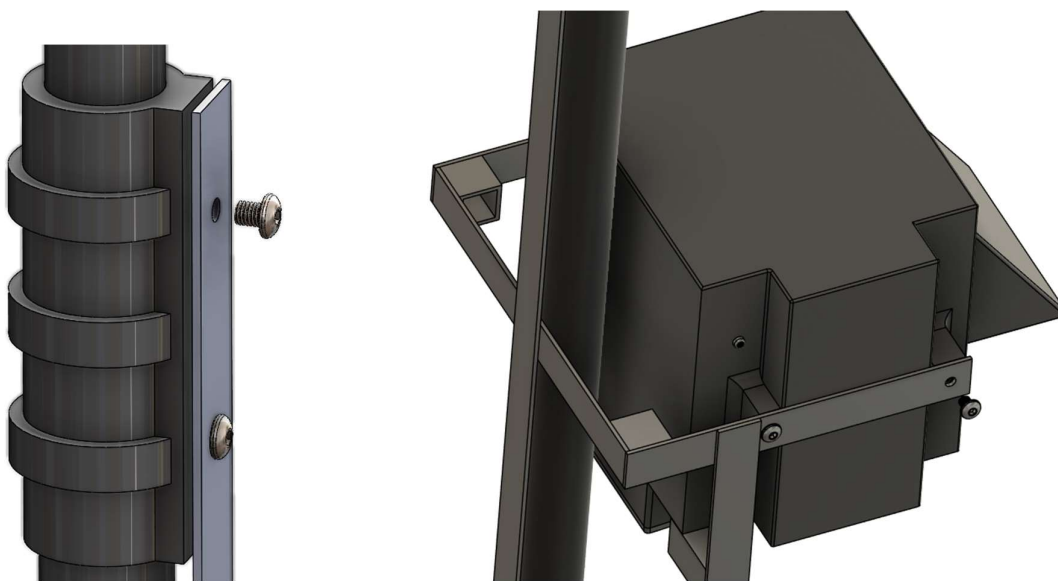


Pole Screws:

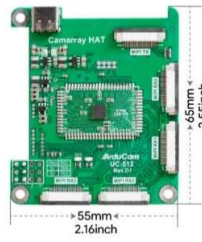
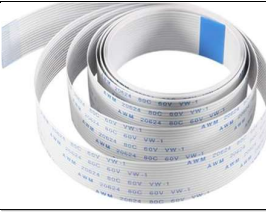
Housing Attachment Nut bolts: <https://www.amazon.com/uxcell-M3x40mm-Stainless-Phillips-Washer/dp/B0789LL18X>

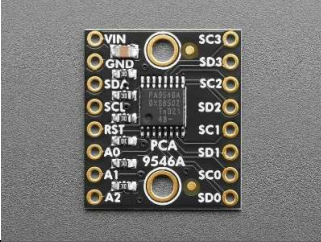


Tripod Knot Screw: <https://www.amazon.com/Buttom-Round-Tensile-Socket-Screws/dp/B07XXNDWKD?th=1>



Components Sourcing

Name	Image	Links
Buck:		https://www.amazon.com/MZHOU-10-72V-Converter-Waterproof-Motorcycle/dp/B0D3V5FZPT?th=1
Start Button:		https://www.amazon.de/-/en/Button-Pressure-Switch-Waterproof-Locking/dp/B08TM3F2NC
LEDs		https://www.amazon.de/-/en/Emitting-Circuits-Projects-Experiments-Multicolor/dp/B07PG84V17
Slider Button		https://www.amazon.de/-/en/sourcing-Vertical-Sliding-Terminals-Locking/dp/B07JGMR3TZ
Camera Mux		https://www.arducam.com/product/arducam-8mp-4-quadrascopic-camera-bundle-kit-b0396/
CSI-2 15– 15 pin Long Wires		https://www.amazon.com/Aokin-Raspberry-Camera-Flexible-Ribbon/dp/B07WF91QML?th=1
CSI-2 – 15-22 pin Wires:		https://www.amazon.com/Flexible-Raspberry-Suitable-Modules-XYGStudy/dp/B0CNV5RJMT?th=1

Camera 15 pin to 15 pin connectors		https://www.reichelt.com/de/en/raspberry-pi-camera-cable-connector-15-pin-to-15-pin-rpi-cam-15-15-p319044.html
i2C MUX		https://www.adafruit.com/product/5663 , https://www.amazon.de/-/en/PCA9546-Channel-I2C-Multiplexer-Compatible/dp/B0BSF13WD7
M2.2x6.5 Screws		https://www.amazon.com/uxcell-M2-2x6-5mm-Phillips-Tapping-Screws/dp/B01EHP3HLC
M3x40 Nut and bolts		https://www.amazon.com/uxcell-M3x40mm-Stainless-Phillips-Washer/dp/B0789LL18X
M5x6 Screws		https://www.amazon.com/Bottom-Round-Tensile-Socket-Screws/dp/B07XXNDWKD?th=1
M3x8 Screws		https://www.amazon.co.uk/ORANXIN-Cross-Head-Tapping-Screws/dp/B084JQ3QT1?th=1
M5 Threaded inserts		https://www.amazon.de/-/en/initeq-M5-0-8-Threaded-Inserts-Printing/dp/B077CG1W3L

Bluetooth Module		https://www.amazon.com/dp/B0C13GVHK?th=1&language=en_US&currency=USD
Jumpers		https://a.co/d/atIbyK4
Type C cable		https://a.co/d/6Rj53DL
20 AWG Wires		https://a.co/d/5hOcLr8
Female headers		https://a.co/d/hAJn5DL
Crimping tool		https://a.co/d/2dkuDHu
PMM20 DC Input and Output		https://amzn.eu/d/d0FMSzg
PMM20 UI Connector		https://amzn.eu/d/ePetRwa